Supplementary information for

Accelerating physics-informed neural networks for full waveform inversion using a hybrid quantum-classical finite-basis architecture

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Supplementary Note: Quantum background

For completeness, we summarize the basic quantum-computing concepts used by the hybrid architecture. A single qubit is a two-level quantum system with state

$$|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle, \qquad |\alpha|^2 + |\beta|^2 = 1, \tag{S1}$$

where $\alpha, \beta \in \mathbb{C}$ are probability amplitudes. An n-qubit register is represented by a normalized statevector in $\mathbb{C}^{2^n}$,

$$|\psi\rangle = \sum_{k=0}^{2^n-1} \alpha_k\,|k\rangle, \qquad \sum_{k=0}^{2^n-1} |\alpha_k|^2 = 1. \tag{S2}$$

The parameterized single-qubit rotations used in this work are

$$R_k(\theta) = e^{-i\theta\,\sigma_k/2}, \qquad k \in \{x, y, z\}, \tag{S3}$$

with explicit matrices

$$R_x(\theta) = \begin{pmatrix} \cos\frac{\theta}{2} & -i\sin\frac{\theta}{2} \\ -i\sin\frac{\theta}{2} & \cos\frac{\theta}{2} \end{pmatrix}, \quad R_y(\theta) = \begin{pmatrix} \cos\frac{\theta}{2} & -\sin\frac{\theta}{2} \\ \sin\frac{\theta}{2} & \cos\frac{\theta}{2} \end{pmatrix}, \quad R_z(\theta) = \begin{pmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{pmatrix}. \tag{S4}$$

The composition $R_z(\theta_3)R_y(\theta_2)R_x(\theta_1)$ spans SU(2) and can prepare any single-qubit pure state up to a physically irrelevant global phase.

Entanglement between neighbouring qubits is introduced by the controlled-NOT (CNOT) gate,

$$\mathrm{CNOT} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \tag{S5}$$

which flips the target qubit if and only if the control qubit is in state $|1\rangle$.

Supplementary Figures

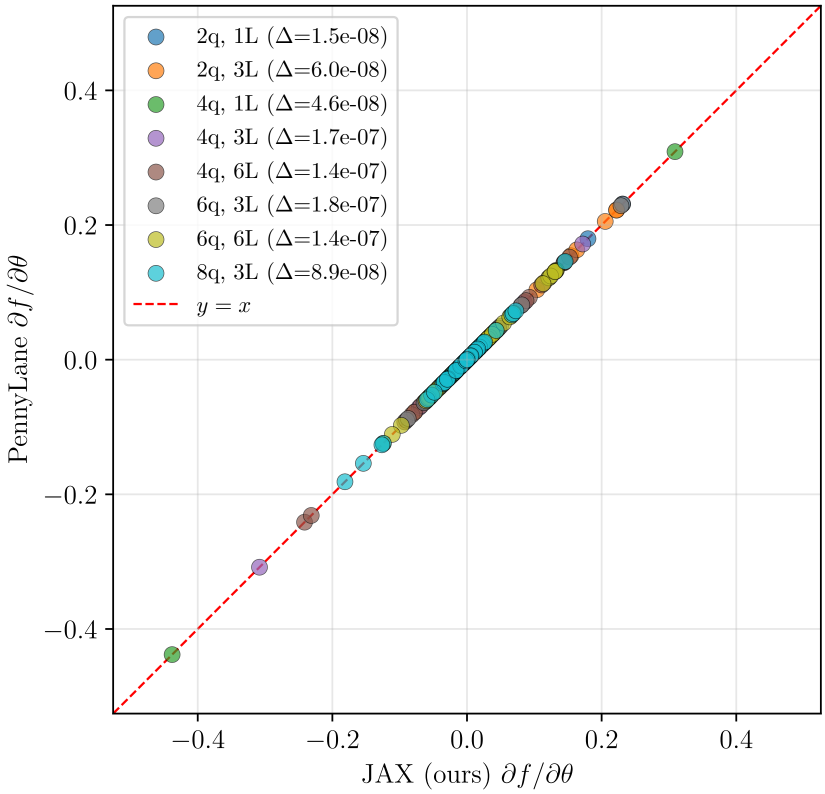

Figure S1: Comparison of parameter gradients $\partial f/\partial\boldsymbol{\theta}_q$ computed by our JAX statevector implementation and PennyLane's `default.qubit` backend across eight circuit configurations (2–8 qubits, 1–6 layers). All values fall on the identity line $y = x$ with maximum absolute deviation $\Delta < 10^{-7}$, confirming numerical equivalence of the two implementations.

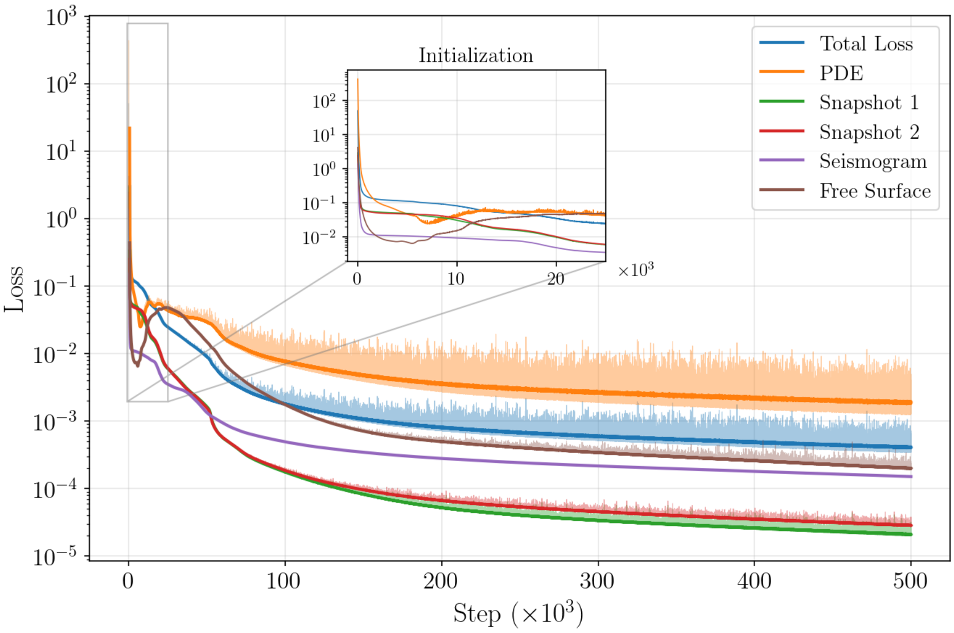

Figure S2: Training loss decomposition for the classical FBPINN baseline on the anomaly benchmark: total loss, PDE residual, displacement-snapshot initial conditions (two components), seismogram misfit, and free-stress boundary condition, plotted versus training iteration. Inset: early-iteration behaviour showing rapid initial loss reduction.

Supplementary Tables

Table S1: Full hyperparameter variant sweep for the anomaly benchmark. $\mathcal{Q}(n_q^{\times n_L})$ denotes a parameterized quantum circuit with n_q qubits and n_L layers. Values marked "–" are identical to the `classical` baseline. Subdomains column shows $\phi(\cdot)$ for wavefield decomposition and $c(\cdot)$ for velocity decomposition.

Model	**ϕ network** $(x,z,t)\to\phi$	**c network** $(x,z)\to c$	**LR**	**Subs**	**Overlap**	$\boldsymbol{\lambda}_{\mathrm{PDE}}$	$\boldsymbol{\lambda}_{\mathrm{IC}}$	$\boldsymbol{\lambda}_{\mathrm{seis}}$	$\boldsymbol{\lambda}_{\mathrm{BC}}$	**# Pars**
`classical`	$\mathcal{N}$: $32^{\times 2}\to 16^{\times 2}$	$\mathcal{N}$: $20^{\times 5}$	10^{-4}	$\phi(5\times3\times5)$	0.35	0.1	1.0	1.0	0.1	151,836
ϕ-network size variants										
`phi16x2`	$\mathcal{N}$: $16^{\times 2}$	–	–	–	–	–	–	–	–	25,836
`phi64x3`	$\mathcal{N}$: $64^{\times 3}$	–	–	–	–	–	–	–	–	649,836
ϕ subdomain decomposition variants										
`sub3x2x3`	–	–	–	$\phi(3\times2\times3)$	–	–	–	–	–	37,779
`sub8x4x8`	–	–	–	$\phi(8\times4\times8)$	–	–	–	–	–	514,017
c-network size variants										
`c10x2`	–	$\mathcal{N}$: $10^{\times 2}$	–	–	–	–	–	–	–	150,226
`c10x3`	–	$\mathcal{N}$: $10^{\times 3}$	–	–	–	–	–	–	–	150,336
`c20x3`	–	$\mathcal{N}$: $20^{\times 3}$	–	–	–	–	–	–	–	150,996
`c40x3`	–	$\mathcal{N}$: $40^{\times 3}$	–	–	–	–	–	–	–	153,516
`c40x5`	–	$\mathcal{N}$: $40^{\times 5}$	–	–	–	–	–	–	–	156,796
`c60x3`	–	$\mathcal{N}$: $60^{\times 3}$	–	–	–	–	–	–	–	157,636
Decomposed c-network variants										
`c_dec5x3`	–	$\mathcal{N}$: $10^{\times 2}$	–	$\phi(5\times3\times5)$ $c(5\times3)$	–	–	–	–	–	152,340
`c_dec8x4`	–	$\mathcal{N}$: $10^{\times 2}$	–	$\phi(5\times3\times5)$ $c(8\times4)$	–	–	–	–	–	154,907
`c_dec5x3_20`	–	$\mathcal{N}$: $20^{\times 2}$	–	$\phi(5\times3\times5)$ $c(5\times3)$	–	–	–	–	–	156,600
Training hyperparameter variants										
`lr5e4`	–	–	5×10^{-4}	–	–	–	–	–	–	–
`lr5e5`	–	–	5×10^{-5}	–	–	–	–	–	–	–
`overlap50`	–	–	–	–	0.50	–	–	–	–	–
`wpde01_wbc01`	–	–	–	–	–	0.01	–	–	0.01	–
`wpde1_wbc1`	–	–	–	–	–	1.0	–	–	1.0	–
Quantum hybrid										
`quantum`	$\mathcal{N}$: $32^{\times 2}$ $\to\mathcal{Q}$: $4^{\times 2}$	$\mathcal{N}$: $20^{\times 3}$ $\to\mathcal{Q}$: $4^{\times 2}$	–	–	–	–	–	–	–	101,812